■ Student Worksheet: Real-Time Music Visualizer & Beat Detector

Name: ____________________________ **Date:** ____________________________ **Class/Period:** ____________________________

Learning Goals

By the end of this activity, you should be able to:

• Explain how computers analyze sound

• Describe what bass, mid, and treble frequencies are

• Understand how code can react to music in real time

• Modify values in code to change program behavior

• Explain how software can control hardware (Arduino LEDs)

Part 1 – How Can a Computer 'See' Music?

Think First: How do you know when a beat drops in a song?

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Key Idea: Sound → Data

Frequency Band	Range (Hz)	Examples
Bass	20–250 Hz	Kick drum, bass guitar
Mid	250–4000 Hz	Vocals, guitar
Treble	4000–16000 Hz	Hi-hats, cymbals

Circle the band you think the human voice mostly lives in: Bass / Mid / Treble

Part 2 – What Is the Program Doing?

What does FFT stand for? ________________________________

What does FFT do to sound? (Circle one)

• Turns sound into colors

• Turns sound into frequencies

• Makes sound louder

• Sends sound to Arduino

Why does this program use threads? (Two things happen at once: listening to sound and drawing the graph.)

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Part 3 – Beat Detection

If the beat sensitivity is too low, what will happen? (Circle one)

• Too many beats detected

• Almost no beats detected

If the beat sensitivity is too high, what will happen? (Circle one)

• Too many beats detected

• Almost no beats detected

Part 4 – Experiment Time

Record the values your class tried:

Setting You Tried	**What Happened?**
Lower threshold	
Higher threshold	
Best setting	

Part 5 – Graphs & Visuals

What do the lines represent? ____

What do the vertical lines mean? ____

Which color was bass? _____________________

Part 6 – Arduino Connection

Beat Type	Number Sent	LED Color
Bass		
Mid		
Treble		

Why do you think the computer sends numbers instead of words?

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Final Reflection

Three things I learned:

1) ____

2) ____

3) ____

One cool idea I have for this tech:

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